

Thesis

Why do these apparently unrelated technical experiences belong in the same career?

01. Thesis

My background includes hands-on mechanical work, construction and fabrication, routed audio systems, paid mixed-system troubleshooting at Level99, and electrical education. I am now making a deliberate transition into robotics, drawing on that experience and subsequent AI-assisted technical project work.

My audio work involved tracing signal paths and routing. A complex audio system is not a collection of independent devices. It is a hierarchy of sources, signal paths, transformations, control signals, interfaces, routing decisions, dependencies, feedback, and consequences. Troubleshooting means understanding the topology well enough to ask the right questions in the right order: Where did the signal originate? What should happen to it next? What transformed it? What controls that transformation? Which path was selected? Where did the observed behavior first diverge from the expected behavior?

Signal-path tracing is a method I can apply to troubleshooting across systems.

Mechanical and automotive work taught me to trace physical causality through engines, drivetrains, electrical faults, and failure modes. Fabrication and welding taught me that a system eventually has to survive material reality: fit, access, sequence, tooling, force, heat, tolerance, repairability, and the difference between what should work on paper and what actually works in the field.

At Level99, my paid Venue Technician work involved troubleshooting and maintaining interactive systems involving RFID and other sensors, PLC-monitored elements, DMX, Arduino hardware, programmable LEDs, audio, mechanical components, and computer control. I used PLC monitoring interfaces to observe states and test sensors; software personnel handled ladder logic. The important lesson was not any one technology. It was the interaction between them. Physical state, sensing, logic, communication, control, and human-visible behavior all had to agree.

At the Starlink facility in Bastrop, Texas, I built clean-room structures for PCB manufacturing. I later completed Martin Electrical School's Journeyman One program, studying electrical theory, code, and safety.

My current technical project work uses AI assistance. I direct and review work, define acceptance criteria, and use tests and recorded evidence to check results. Signal-path reasoning is useful here for examining inputs, transformations, dependencies, and failures; the project artifacts show what has actually been checked.

These are methods I want to bring to robotics build and test work.

02. Why robotics

Robotics is compelling to me because it compresses nearly every one of those domains into the same machine. A robot is simultaneously a mechanical structure, an electrical system, a collection of sensors and signal paths, a control architecture, a software system, a communications system, a set of interfaces between abstractions, and a physical actor whose outputs change the real world.

I am choosing robotics as my next direction because it brings these systems together. My existing experience can support build, maintenance, and troubleshooting work while I develop the robotics-specific knowledge I still need.

03. How I tend to model systems

Vocabulary

Signal - Transformation - Routing - Control - Interface - Feedback - Consequence

A signal might be audio, voltage, sensor data, a software event, a model output, or a human instruction. A transformation might be amplification, mechanical transmission, computation, interpretation, or policy evaluation. Routing determines where something goes and therefore what can happen next. Control determines which transformations or routes are permitted, selected, or inhibited. Interfaces are where assumptions meet and where meaning is frequently changed, lost, or misinterpreted. Feedback tells the system - or the operator - what actually happened. Consequence is the final reminder that eventually a technical system does something observable, and in robotics that consequence can be physical.

Background informing my robotics transition: Mechanical troubleshooting; construction / fabrication; routed audio systems; Level99 troubleshooting; electrical education; AI-assisted project work; current robotics transition.

04. What I am bringing into robotics

I am not presenting myself as an experienced robotics engineer, embedded-firmware specialist, controls engineer, or someone with years of professional autonomy development. What I can bring immediately is a different combination: hands-on fabrication and welding experience; mechanical and automotive troubleshooting; experience using tools and equipment in real physical environments; electrical fundamentals and formal electrical study; sensor and interactive-system troubleshooting experience; deep familiarity with signal-path reasoning from audio engineering; software and AI systems thinking; comfort learning across domain boundaries; and a strong bias toward tracing causes rather than treating symptoms.

That combination is still developing. That is part of why I am at RoboBoston. I am looking for environments where I can contribute at the level I have actually earned while continuing to move deeper into robotics, electromechanical systems, integration, test, reliability, controls, and the architecture of systems that have to work outside of a diagram.

05. The general thesis

I now bring together mechanical troubleshooting, construction and fabrication, routed audio work, Level99 maintenance, and electrical education as I pursue robotics technician work. My subsequent AI-assisted projects add work on requirements, tests, and evidence, with their limits stated alongside the results.

Robotics is where all of those layers become consequential at once. That is the work I am trying to move toward.